

Exercises

Exercise 4: Cleaning, summarising and linking data

Read Chapter 3 to help you complete the questions in this exercise.

This exercise picks up exactly where Exercise 3 left off and uses the same `cardiac` dataframe. If you are starting a fresh R session, either continue with the R script you wrote for Exercise 3 or run the two lines below before you go any further. They are the import from Exercise 3, Q5 and the two factors from Exercise 3, Q7.

```
cardiac <- read.table('data/cardiacdata.txt', header = TRUE, sep = "\t", stringsAsFactors = TRUE)

cardiac$Fsex <- factor(cardiac$sex, levels = c(1, 2),
                      labels = c("Female", "Male"))

cardiac$Fsmoking <- factor(cardiac$smoking, levels = c(1, 2, 3),
                          labels = c("Current", "Ex", "Never"))
```

1. In Exercise 3 you learned how to pick out rows that meet a condition. You can use that skill to check whether your data are believable. This matters a lot: a value that is simply wrong will not stop your code running, will not produce a warning, and will quietly change every result you calculate from it. Run `summary()` on your dataframe again and look at the minimum and maximum of each numeric variable. Three variables contain values that are not merely unusual but impossible. Two are impossibly low and one is impossibly high (hint: a body mass index below about 15 or above about 60 is essentially unheard of in a living adult, and nobody can survive with no cholesterol in their blood at all, so a blood fat concentration of exactly 0 is not a low reading, it is an impossible one). Find them, work out which patient the impossibly high one belongs to, and then set all of the offending values to `NA` so they cannot contaminate anything you calculate later (you will need the square bracket `[ ]` notation from Section 3.4.2, this time on the left hand side of an assignment). Re-run `summary()` afterwards to check it worked. Why is `NA` the right answer here, rather than deleting the whole patient record or guessing what the value should have been?

2. Another useful way to manipulate your dataframes is to sort the rows based on the value of a variable, or on a combination of variables. Rather counter-intuitively you should use the `order()` function to sort your dataframes, not the `sort()` function (see Section 3.4.3 of the Introduction to R book for an explanation). Ordering dataframes uses the same logic you practised in Q12 in Exercise 2. Sort all rows in the `cardiac` dataframe by ascending order of systolic pressure within each level of smoking status, and assign the result to a variable with a sensible name. The trick is to remember that `order()` will take more than one variable, and that the order you give them in matters. Now take a look at the bottom of your sorted dataframe. Where have the patients with a missing smoking status ended up, and why?

3. Often, we would like to summarise variables by, for example, calculating a mean, median or counting the number of observations. To do this for a single variable it's fairly straight forward :

```
mean(cardiac$age)           # mean age
median(cardiac$systolic)    # median systolic blood pressure
length(cardiac$tchol)      # number of observations
```

3. (cont) Perhaps more interestingly, you might want to summarise one variable conditional on the level of another categorical variable, and to do several variables at once. The `aggregate()` function does exactly this (see Section 3.5 of the Introduction to R book, or `?aggregate`). Use `aggregate()` to calculate the mean age, systolic pressure, diastolic pressure and total cholesterol for each smoking category. Then calculate the same means for each combination of smoking category and sex.

The grouping variables go inside the `by` argument wrapped in `list()`, because a list is how `aggregate()` accepts more than one of them at once.

A word of warning before you start. Remember those missing values you have been tripping over since Exercise 3? They have not gone away, and `tchol` still has two of them. Run your first `aggregate()` without any special handling, look hard at the `tchol` column, and only then reach for the `na.rm` argument.

4. Knowing how many observations are present for each category (or combinations of categories) is useful to determine whether you have an adequate sample size (for subsequent modelling for example). Use the `table()` function to determine the number of patients in each smoking category (see Section 3.5 again for more information). Next use the same function to display the number of patients for each combination of smoking category and sex. Does `table()` tell you about the patients whose smoking status is missing?

5. Not every variable is symmetric, and several of the ones here are not. Look back at `triglyceride` in your `summary()` output. The mean sits noticeably above the median, which is the signature of a long right hand tail, a few patients with much higher values than everybody else pulling the average up. Transforming to a log scale pulls that tail in, and it is routine practice for blood lipids.

Create a new variable in the `cardiac` dataframe called `log_triglyceride`, holding the base 10 logarithm of `triglyceride` (see `?log10`, and Section 3.4.4 for adding a column). Compare the mean and the median before and after. What has happened to the gap between them?

Data linkage

Real analyses almost never involve a single file. The measurements you want are usually spread across several sources and you have to bring them together first. When those sources hold records about the same people, combining them into one dataset is called data linkage, and it is an important technique in health data science. Hospital admissions linked to prescribing records, a birth cohort linked to school attainment, a patient list linked to the death register. Each linkage answers a question that neither source could answer on its own, and this is the term we will use for it throughout the course.

The idea is simple. You match records on something that identifies the same person in both sources, usually an identifier such as a patient number. However, if implemented carelessly, analyses can go quietly wrong,

which is what the next three questions are about. In R the function that does the work is `merge()` (see Section 3.4.5 of the Introduction to R book), and the operation it performs is called a join.

6. The patients in this study were followed up ten years after their first examination, and those follow-up measurements are held in a separate file. Download '*cardiac_followup.txt*' from the **Data** link, save it to your **data** directory and import it with `read.table()` into a variable called **followup**. How many rows does it have, and how does that compare with **cardiac**?

Now use the `merge()` function to link the two together on the patient number, and assign the result to **cardiac_fu**. Left to its own devices `merge()` performs what is called an inner join, which keeps only the patients who appear in both sources. How many rows does the linked dataframe have, and can you explain why?

7. Losing 55 patients without being told is exactly the sort of thing that can ruin an analysis, and it is why you check row counts every single time you link two sources. Often what you actually want is a left join, which keeps every patient from the first dataframe whether or not they have a match in the second. Look at Section 3.4.5 and the help file for `merge()`, and find the argument that switches `merge()` from an inner join to a left join. Redo the linkage keeping all 163 patients, assign it to **cardiac_all**, and check how many of them have a missing follow-up systolic pressure.

8. So far the two files you have linked have both been one row per patient, which is the easy case. Real linkage is usually messier than that, because the thing you are linking to happens more than once.

Download '*cardiac_admissions.txt*' from the **Data** link, save it to your **data** directory and import it into a variable called **admissions**. This file holds one row per **hospital admission** during the ten year follow up, not one row per patient. Note that unlike the rest of the cardiac data, which is real, these admission records are simulated.

Use `str()` and `nrow()` to get your bearings. How many admissions are there, and how many different patients do they belong to? Then, before you join anything, check the key in both directions. Are there any patients in **cardiac** with no admissions at all? Are there any admissions belonging to a patient number that is not in **cardiac**, and if so, how would that happen?

Now join **cardiac** to **admissions** with `merge()` as you did in Q6, and look at the number of rows. Compare `mean(age)` before and after the join. Something has gone wrong. What?

9. So the join in Q8 is not the one you wanted. What you actually want is to keep one row per patient and attach a *summary* of that patient's admissions, and to do that you have to summarise the admissions **before** you link them, not after.

- a) Use `aggregate()`, as you did in Q3, to produce one row per patient from **admissions**, holding the number of admissions and the total number of days spent in hospital. Counting rows per group is what `FUN = length` is for, and total bed days is `FUN = sum` applied to `los`.
- b) Now link that summary onto **cardiac**, keeping all 163 patients as you did in Q7. Check the row count.
- c) Look at the patients who were never admitted. What has the join put in their count column, and is that the right value? This is the one place in this whole course where an **NA** does not mean 'we do not know'. Work out what it does mean here, decide what to do about it, and calculate the mean number of admissions per patient both ways to see how much the decision matters.
- d) Finally, the question the linkage was for. Does the number of admissions differ by smoking status?

10. Ok, we have spent quite a bit of time (and energy) learning how to clean, summarise and link dataframes. The last thing we need to cover is how to export a dataframe from R to an external file (see Section 3.6 of the book for more details). Export your cleaned and linked dataframe `cardiac_all` to a file called 'cardiac_clean.txt' in the `output` directory you created in Exercise 1. To do this you will need to use the `write.table()` function. You want to include the variable names in the first row of the file, but you don't want to include the row names. Also, make sure the file is a tab delimited file. Once you have created your file, try to open it in Microsoft Excel (or open source equivalent). Finally, and this is the part people forget: add a comment block at the top of your R script listing every change you made to the data and why. Which values did you set to `NA`, in which variables, and which patients did they belong to? Someone reading your work in six months, including you, needs to be able to answer that without re-running anything.

End of Exercise 4